PI7C9X3G1632GPF / PI7C9X3G1224GPB / PI7C9X3G816GPC / PI7C9X3G808GPC / PI7C9X3G606GPC PCI Express Gen 3 Packet Switch: Errata

INTRODUCTION

This Errata document is created based upon the compatibility test results of PI7C9X3G1632GPF / PI7C9X3G1224GPB / PI7C9X3G816GPC / PI7C9X3G808GPC / PI7C9X3G606GPC.

DOCUMENT REVISION HISTORY

DATE	REVISION	DESCRIPTION
09/07/2023	0.1	New Errata document
01/11/2024	0.2	Added E3 and E4
05/22/2024	0.3	Added E5
06/11/2024	0.4	Added E6
07/16/2024	0.5	Updated “Date” in Silicon Revision Section
11/25/2024	0.6	Added E7, E8 and E9
12/11/2024	0.7	Added E10
03/07/2025	0.8	Added E11
04/09/2025	0.9	Added E12
08/09/2025	0.a	Added E13
10/16/2025	0.b	Added E14

SILICON REVISION

DATE	REVISION ID	DESCRIPTION
08/28/2023	0Fh	PI7C9X3G1632GPF / PI7C9X3G816GPC
09/07/2023	0Fh	PI7C9X3G808GPC
09/27/2023	0Fh	PI7C9X3G1224GPB
10/11/2023	0Fh	PI7C9X3G606GPC

SUMMARY OF ERRATA

ERRATA #	DESCRIPTION	STATUS
E1	Not Support Vendor_defined Type 1 Message Which Length Is Not 0 or 1	Workaround available
E2	False Bad DLLP Error at GEN3 Port 0	Workaround available
E3	Earlier Deassert INT Message When INT Signal Shared By Multiple Ports	Workaround available
E4	In a Credit Deficit Condition May Occur Receiver Overflow and Flow Control Protocol Errors	Workaround available
E5	PI7C9X3G1632GPF Incompatible with AMD Platform When Using x16 Link Width	No workaround available
E6	Link Failure During the Second Link Speed Change from Gen1/2 to Gen3 at Upstream Port	Workaround available
E7	Compatibility Issue Due to ASPM L1	Workaround available
E8	I2C Reads Occasionally Return All 0's	Workaround available
E9	Power Saving Not Functional When Lane Reversible Is Implemented	Workaround available
E10	PI7C9X3G1224GPB May Be Struck Due to Full Internal Buffer	Workaround available
E11	System Hang when DPC Event Is Triggered	Workaround available
E12	Not Support Hot Plug Interrupt Triggered by "Command Completed" Event	Workaround available
E13	System Hang-Up Due to False "No Data Credit Available" Alert	Workaround available
E14	Violate SMBUS Read Data Hold Time Spec.	No workaround available

E1: Not Support Vendor_defined Type 1 Message Which Length Is not 0 or 1

Problem:
If the RC or EP sends out a Vendor_defined type 1 message whose length is not 0 or 1, the switch will consider it a malformed packet and drop the message.

Workaround:
The RC or EP can limit the lengths of Vendor_defined type 1 messages they send out to 0 or 1.

Status:
No fixes for this issue are planned.

E2: False Bad DLLP Error at GEN3 Port 0

Problem:
The switch port 0 may falsely report a bad dllp error at the Correctable Error Status Register when the link is GEN3 speed. In GEN3 speed, TLP and DLLP are faming by token. If some TLP packets which data payload inside a DLLP-like token, even switch can filter out such fake DLLP, but still do crc16 check for this fake DLLP packet and got failed result.

Workaround:
Disable switch to report bad dllp error by eeprom (set offset 43Ch.bit[9]=0).

Status:
No fixes for this issue are planned.

E3: Earlier Deassert INT Message When INT Signal Shared By Multiple Ports

Problem:
The PCIe packet switch will forward the de-asserted INT message directly once receive. It may cause CPU can not service some of the endpoints which share the same INT message.

The following scenario shows the sequence of the event. For example, after routed by the PCIe switch, two endpoints (EP1 and EP2) share INTA as their interrupt signal. EP1 and EP2 Assert INT message at the same time and the PCIe switch forwards the Assert INTA messages to CPU. The CPU receives the INT messages, and services EP1's INT message first. Then EP1 sends out De-asserted INT message and the PCIe switch will forward to CPU. Then CPU will stop to service. If it is earlier than CPU services EP2's INT message, it will cause EP2's INT message is not serviced. As a result, the PCI will not service EP2 anymore as EP2 keeps in Assert INT state.

Workaround:
There have two solutions,

1. When BIOS/OS receive an Assert INT message, BIOS can keep to service all EPs which share the same INT message, rather than stop to service when receive Deassert INT message, it can avoid this issue.
2. If your system has INTA/B/C/D in-band signals, you can OR INT pins together which share the same INT after routed by the PCIe switch.

Status:
No fixes for this issue are planned.

E4: In a Credit Deficit Condition May Occur Receiver Overflow and Flow Control Protocol Errors

Problem:
When forwarding data Post Write packets with turning on Relax Ordering (RO) bit in a credit deficit condition, the switch port may report Flow Control Protocol Error (FCPE) while Root Complex or Endpoint assert Receiver Overflow via Advanced Error Register. FCPE is belong to fatal error message which sometimes results into system hang-up. On the other hand, the receiver overflow will cause application execution error as well.

PCIe utilizes flow-control credit based protocol to regulate traffic not overflowing the input data queue. The packets generated from the egress port of packet switch must ensure the credits consumed by the transmitted packet is less than or equal to the available credits posted by the ingress port of Root Complex or Endpoint. Otherwise, their input data queue has a chance to be overflow. Meanwhile, the flow control credit calculation in switch port will violate protocol rule to flag FCPE as credit consumed exceeds credit limit.

The switch port data buffer is constructed by dual queues. When the 1st queue is full, the packet will be stored in the 2nd queue. In this use case, the 1st queue is always full as the credit update is slow from the destination port (i.e. Root Complex). When both of two queues raise requests for packet delivery service, a grant may go to the wrong queue if RO turned on to use incorrect consumed credit. This will mess up flow control credit calculation under the situation of available credit in deficiency.

Workaround:
This error can be workaround by preventing from both of queues to make assertion at the same time. Therefore, the grant can serve the corresponding request with a correct consumed credit to ensure the integrity of flow control credit calculation.

Please turn on the bit[14] of TL CSR #1 register, which is at offset 0x4C4. This will block the post packet request from one queue passing over other post packet request from another queue, so the request from the queues is one at a time.

Status:
No fixes for this issue are planned.

E5: PI7C9X3G1632GPF Incompatible with AMD Platform When Using x16 Link Width

Problem:
The PI7C9X3G1632GPF packet switch sends out back-to-back (no gap) InitFC packets in DL_Init (Initial Flow Control) state. On AMD platform, packet switch works without any issues if the link width is x1, x2, x4 or x8. However, if the packet switch uses x16 link width, the DL_Init state can not be completed. As a result, the packet switch can not send out any TLP packets, and the system can not detect the PI7C9X3G1632GPF packet switch.

Workaround:
No workaround available.

Status:
No fixes for this issue are planned.

E6: Link Failure During the Second Link Speed Change from Gen1/2 to Gen3 at Upstream Port

Problem:
After the initial power-on is completed, the packet switch performs a speed change from GEN1 to GEN3 if the link partner at Upstream Port is capable of GEN3 speed. Because the packet switch does not save coefficient result obtained from the first link speed change, the second speed change will fail if the software tries to re-train the link from GEN1 or GEN2 to GEN3 again. As a result, the link at Upstream Port will fail.

Workaround:
Set the “Performance Equalization” bit in the Link Control 3 Register at root complex.

Status:
No fixes for this issue are planned.

E7: Compatibility Issue Due to ASPM L1

Problem:
When the ASPM L1 function is enabled by the BIOS/OS, compatibility issue may occur and the system may experience the problem of link-down.

Workaround:
Set bit[11:10]=00b of Link Capability register at offset 0x74 to disable ASPM L1 capability via EEPROM.

Status:
No fixes for this issue are planned.

E8: I2C Reads Occasionally Return All 0's

Problem:
The Packet switch has an internal ring arbiter to handle all CFG packets, I2C/SMBUS packets and Messages generated by switch itself. When those packets are coming at the same time, the arbiter may hang occasionally, and as the result, I2C reads will return all 0s.

Workaround:
Set bit[2]=1 of Port 0's (Tile 0 and Tile 1 if two-tile switch) TL CSR #3 register at offset 0x4CC to force the internal ring arbiter to handle packets one at a time to avoid this issue.

Status:
No fixes for this issue are planned.

E9: Power Saving Not Functional When Lane Reversible Is Implemented

Problem:
When Lane Reversible is implemented, power saving function can not be enabled as it will cause link-down problem after the link returns from L1 state.

Workaround:
The power saving function is disabled by default. Be sure not to enable the function if the port has Lane Reversible implemented.

Status:
No fixes for this issue are planned.

E10: PI7C9X3G1224GPB May Hang Due to Full Internal Buffer

Problem:
The flow control credit update of PI7C9X1224GPB at Tile1 P0 may not operate correctly under certain condition when it is configured as x8 lane width which Port Configuration is set as 000b. The Switch may hang when the link partner updates the header credit with a number larger than 3 in one transaction under the credit deficit situation. This update may not reflect in the available credit counter. As a result, the Switch will not transmit all the packets, and the buffer is not released. Eventually, packets from the root port cannot enter into the upstream port of the Switch anymore due to the full internal buffer.

Workaround:
1. Avoid setting the Port Configuration at Tile 1 to 000b. That is, the Downstream port at Tile 1 can not be x8 lane width.
2. If the application configures the PI7C9X1224GP as three x8 ports, select Tile 1, Port 0 as the Upstream Port.

Status:
No fixes for this issue are planned.

E11: System Hang when DPC Event Is Triggered

Problem:
When DPC function is enabled and DPC Completion Control bit is set to 0b, switch will response completion packet with CA (completion abort) status when DPC event is triggered which will cause switch hang internally. As the end, system will hang.

Workaround:
When DPC function is enabled, set DPC Completion Control bit to 1b, switch will response completion packet with UR (unsupported request) status when DPC event is triggered. It can avoid this hang issue.

Status:
No fixes for this issue are planned.

E12: Not Support Hot Plug Interrupt Triggered by “Command Completed” Event

Problem:
When Hot Plug interrupt is triggered by “Command Completed” event, it will cause repetitive interrupts as “Command Completed” status can not be clear. As the result, system will hang.

Workaround:
Set “No Command Completed Support” bit in Slot Capability register by eeprom to avoid this issue.

Status:
No fixes for this issue are planned.

E13: System Hang-Up Due to False “No Data Credit Available” Alert

Problem:
This issue occurs when the switch needs to generate data packets in response of the following requests such as Set_Slot_Power message, MSI/MSI-X message and non-post read to CDLEP port from a remote host.

In a rare condition, the above-mentioned requests just hit transmission not ready. Then the calculation of the available data credits will be incorrect. The calculated available data credits will be one more credit than the credits the host advises during initial flow control. This discrepancy of one data credit will accumulate until the switch’s credit counter is overrun (from 4095 to 0). At this point, there will be no available data credits to return the last data packet, and the system will hang as a result.

Workaround:
1. Remove slot capability or MSI/MSI-X capability by EEPROM
2. Remote host does not issue any read command to CDLEP port responding with DATA which can use I2C read command instead

Status:
No fixes for this issue are planned.

E14: Violate SMBUS Read Data Hold Time Spec.

Problem:
Switch’s smbus read data hold time violate spec. which need be >= 300 ns). The data hold time for switch is about 100 ns.

Workaround:
Use I2C to instead SMBUS.

Status:
No fixes for this issue are planned.